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Gradient Decomposition and Alignment for Incremental Object Detection

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Abstract

Incremental object detection (IOD) is crucial for enabling AI systems to continuously learn new object classes over time while retaining knowledge of previously learned categories, allowing model to adapt to dynamic environments without forgetting prior information. Existing IOD methods primarily employ knowledge distillation to mitigate catastrophic forgetting, yet these approaches overlook class overlap issues, often resulting in suboptimal performance. In this paper, we propose a novel framework for IOD that leverages a decoupled gradient alignment technique on top of the specially proposed pseudo-labeling strategy. Our method employs a Gaussian Mixture Model to accurately estimate pseudo-labels of previously learned objects in current training images, effectively functioning as a knowledge-replay mechanism. This strategy reinforces prior knowledge retention and prevents the misclassification of unannotated foreground objects from earlier classes as background. Furthermore, we introduce an adaptive gradient decomposition and alignment method to maintain model stability while facilitating positive knowledge transfer. By aligning gradients from both old and new classes, our approach preserves previously learned knowledge while enhancing plasticity for new tasks. Extensive experiments on two IOD benchmarks demonstrate the effectiveness of the proposed method, achieving superior performances to state-of-the-art methods. The code and datasets are available at https://github.com/FHR-L/GDA-IOD.

1. Introduction

Object detection is a fundamental computer vision task, driving progress in numerous specialized areas. Existing general object detection models [9, 10, 40, 42] assume access to all target classes and fully labeled data from the start, with a fixed data distribution throughout training. However, in real-world scenarios, new object categories often appear over time. This dynamic setting requires object detection models to continuously adapt to these new categories, while also addressing challenges like limited storage for all past data and data privacy concerns. To address this, incremental object detection (IOD) [38] has thus emerged as an essential approach, inspired by biological visual systems that continuously acquire, integrate, and reorganize knowledge. IOD enables models to recognize new objects as they appear without forgetting knowledge of learned categories.

The existing common way to solve IOD is through knowledge distillation [8, 11, 14, 27, 32, 34, 55]. Researches have shown that using feature/prediction distillation can significantly reduce catastrophic forgetting in these models. In one implementation [11], incremental learning was applied to an anchor-free detector, where knowledge distillation selectively focused on non-regression outputs, effectively preserving previously learned knowledge while adapting to new classes. However, most methods overlook the unique challenge in IOD, foreground objects identified in earlier stages are left unlabeled in later stages, which can cause the model to misclassify them as background. While distillation-based approaches emphasize strong regularization to prevent catastrophic forgetting, they fail to address class overlap, leading to suboptimal performance on both previously learned and new classes.

To address the challenges of *catastrophic forgetting* in incremental object detection, we propose a comprehensive framework featuring a novel gradient decomposition and alignment method. This method is built on an advanced pseudo-labeling strategy to address missing annotations of previously learned class objects in current training images, effectively serving as a knowledge replay mechanism. Specifically, it employs a Gaussian Mixture model to accurately estimate pseudo-labels and applies adaptive gradient alignment with these previously learned class objects, enabling parameter updates that preserve prior knowledge while promoting positive backward transfer.

Specifically, accurately estimating the pseudo old-class

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objects in current training images has the following two merits: (1) it acts as a knowledge-replay mechanism, reinforcing retention of prior knowledge for anti-forgetting, and (2) it helps prevent the model from misclassifying unannotated foreground objects from previous classes as background, avoiding severer knowledge forgetting. To achieve this, we introduce a Gaussian Mixture model that accurately estimates pseudo-labels for overlapping object classes from previous tasks in the current training images. Unlike previous methods [29, 32] that rely on selecting only the highest-confidence predictions from the old model, our approach leverages the common bimodal distribution observed in inference results (As shown in Fig.1), where medium-confidence predictions often indicate those correctly detected but misclassified foreground instances. Using a Gaussian Mixture Model (GMM) with two components, we segment predictions adaptively into low-, medium-, and high-confidence regions. Each region is then treated accordingly: high-confidence predictions reinforce knowledge of prior classes, while medium-confidence ones are recognized as potential foreground targets, excluded from negative samples to prevent the Region Proposal Network (RPN) from misclassifying them as background. This approach maximizes the use of pseudo-label information while reducing noise, enhancing both knowledge retention and detection accuracy.

Building on estimated pseudo-labels for previously learned object classes and current annotations for balanced model training, we propose an adaptive gradient decomposition and alignment method for incremental object detection. This approach achieves a dynamic balance between model stability and plasticity by adjusting parameter gradients from both old-class objects and new training samples, guided by their influence on the loss function. The core idea is to ensure that parameter updates for the new classes do not negatively impact knowledge from previously learned classes, represented by the gradient from old-class objects. Specifically, at each iteration, the gradient is decomposed into two parts: one from old-class objects and one from new training objects. When these gradients align (with an angle under 90 degrees), the update direction facilitates learning from new classes while positively transferring knowledge to prior tasks. However, if the gradients diverge (angle above 90 degrees), the update direction is adjusted orthogonally to the old-class gradient, preserving stability without disrupting prior knowledge. This mechanism enhances model stability while fostering beneficial knowledge transfer.

The main contributions can be summarized as follows:

- To tackle the challenging IOD task, we introduce a Gaussian Mixture Model to accurately estimate pseudo-labels for old-class objects in current training images. Such strategy serves as knowledge replay to reinforce prior knowledge retention and prevent misclassifying unannotated foreground objects from previous classes as background, greatly reducing knowledge forgetting.
- Building on the estimated old-class objects and current training data, we devise an adaptive gradient alignment method for IOD, which enhances model stability while fostering beneficial knowledge transfer for previous object classes under the guidance of gradient decomposition mechanism.
- Extensive experimental results demonstrate the effectiveness of anti-forgetting on two incremental object detection benchmarks with diverse experiment settings, and our approach achieves superior performances to state-of-the-art methods.

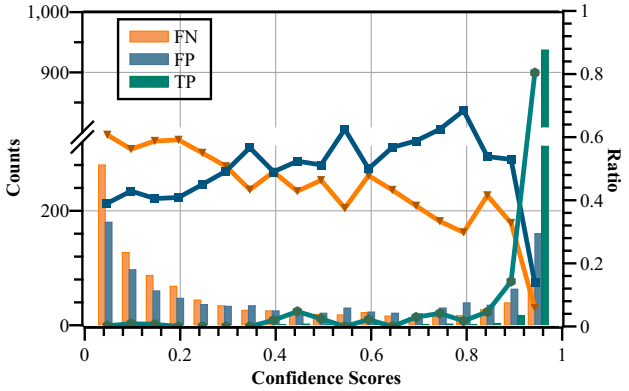


Figure 1. Visualization of bimodal distribution of $\mathcal{M}_{t-1}$'s inference results on $\mathcal{D}_t$ under VOC 10-10 scenario. TP, FP, and FN represent the true positive, false positive, and false negative (IoU at 0.8), respectively. Surpassing half of the prediction in medium-confidence is foreground object but misclassified (False Positive).

2. Related Work

Object Detection. Traditional object detection models have evolved into two main paradigms: single-stage detectors [25, 26, 40, 41] and multi-stage detectors [12, 16, 24, 35]. Single-stage detectors are designed to be computationally efficient and suitable for real-time applications, but often trade-off some accuracy. Faster R-CNN [35] is a significant milestone for multi-stage detectors, which introduces a Region Proposal Network (RPN) that enables end-to-end training for both region proposal generation and object classification. In this work, we focus on extending the two-stage detector, Faster RCNN, to an incremental learning paradigm where the model needs to learn categories in multiple steps and maintain previous knowledge.

Class Incremental Learning. Continual learning aims to address the challenge of enabling models to learn from a continuous stream of data while mitigating catastrophic forgetting, where performance on previously learned tasks degrades as the model learns new information. Incremental classification has attracted much attention in recent years. Existing methods can be broadly categorized

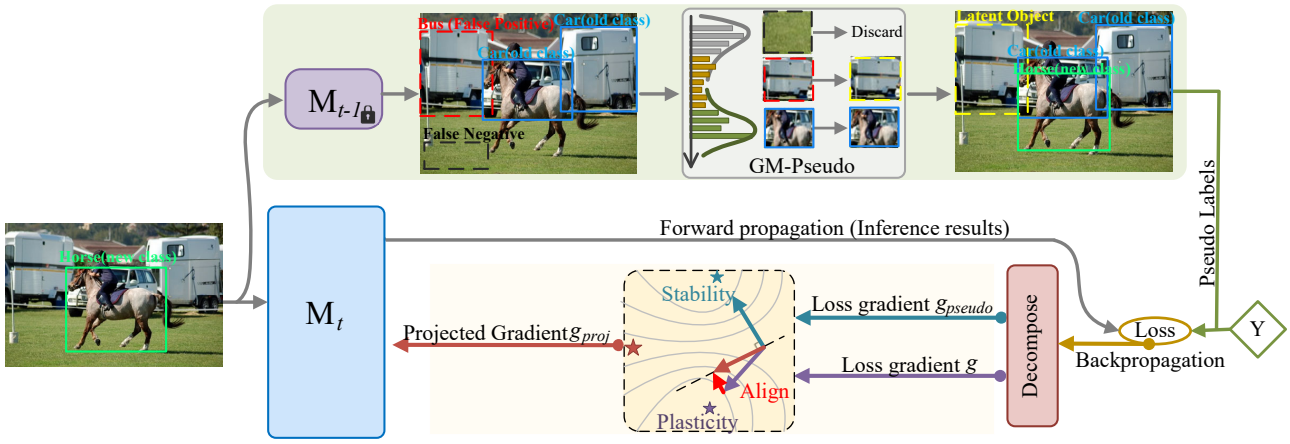


Figure 2. Overall framework of our proposed **GDA-IOD** for Incremental Object Detection. The unannotated old class instances are relabeled by $\mathcal{M}_{t-1}$ and adaptively segment into low-, medium-, and high-confidence parts by GM-Pseudo module. We select the high-confidence part as pseudo labels for supervision training and exclude the medium-confidence part from being misclassified as background. During optimization, we decompose the gradient from old-class objects as $\boldsymbol{g}_{pseudo}$, align overall gradients $\boldsymbol{g}$ on the orthogonal direction of $\boldsymbol{g}_{pseudo}$ to ensure parameter updating does not negatively impact learned knowledge.

into regularization-based [1, 2, 21, 22, 30, 31, 52], replay-based [3, 5, 6, 17, 18, 37, 53], and dynamic architecture approaches [33, 36, 39, 44, 45, 47, 51, 54]. Specifically, EWC [21] introduces constraints on model parameters to preserve knowledge. [5] replay examples with high prediction entropy and near the decision boundary. PNN [36] isolates previous knowledge from interference by expanding the model's architecture, learning a new backbone for each task.

Incremental Object Detection. Compared with incremental classification, incremental object detection presents additional challenges due to the need to retain spatial and semantic information about previously seen objects. ILOD [38] is the pioneer in proposing the first incremental object detection method based on Fast RCNN [12] and adapting knowledge distillation to transfer the old knowledge to the new model. Inspired by this work, some following methods [8, 11, 14, 27, 32, 55] extend the distillation framework to Faster RCNN. AFD [27] proposes an attentive feature distillation framework to leverage dual-direction attention distillation to mitigate forgetting in incremental detection. [8] proposes a dual-teacher distillation framework and the use of in-the-wild data to bridge the non-co-occurrence of base classes in the training data of the novel classes. Replay-based methods [28, 29, 46] have shown more promise, where a subset of past examples or synthetic data is stored and periodically replayed to reinforce prior knowledge. ABR [28] uses mixup and mosaic box augmentation strategies to replay previous objects as an alternative to image replay for training.

In this work, we focus on solving catastrophic forgetting by adaptive pseudo-labeling strategies with gradient-based parameter regularization *without example replay*.

3. Approach

3.1. Preliminaries

Problem Formulation. Traditional object detection assumes that the training dataset $\mathcal{D}$ includes a fixed set of object classes $\mathcal{C}$, which are available all at once from the beginning. In contrast, incremental object detection involves training over N sequential stages($N > 1$), where the model encounters a new set of classes at each stage. Let $\mathcal{T} = \{\mathcal{T}_1, ..., \mathcal{T}_N\}$ represent these N incremental learning stages. In each stage $\mathcal{T}_t$, the model has access only to dataset $\mathcal{D}_t \subset \mathcal{D}$, which includes annotations $\mathcal{Y}_t$ for object instances in a subset of categories $\mathcal{C}_t \subset \mathcal{C}$. These subsets are disjoint across stages, meaning $\mathcal{C}_n \cap \mathcal{C}_m = \emptyset$ for any $n \neq m$ and $n, m < N$. The model, denoted as $\mathcal{M}$, can be updated from $\mathcal{M}_{t-1}$ to $\mathcal{M}_t$ using only $\mathcal{D}_t$ at each stage $\mathcal{T}_t$, without revisiting previous datasets, while retaining knowledge of previously learned categories. The primary challenge is ensuring that $\mathcal{M}_t$ continues to perform well across all previously learned classes, $\cup_{i=1}^{t-1}\mathcal{C}_i$.

General Object Detector. Following [4, 28, 32, 50], our work focuses on detection models within the R-CNN series, specifically implementing Faster R-CNN [35]. Faster R-CNN is an end-to-end model comprising three main components: a feature extractor $\mathcal{M}_f$, a Region Proposal Network (RPN) $\mathcal{M}_{rpn}$, and a detector head (ROI head) $\mathcal{M}_{roi}$. The feature extractor $\mathcal{M}_f$ processes the input image to generate a dense feature map. Then, the RPN $\mathcal{M}_{rpn}$ predicts bounding box offsets $\boldsymbol{l} = \{l_x, l_y, l_w, l_h\}$ and objectness scores p for region proposals based on predefined anchors at each spatial location. The detector head $\mathcal{M}_{roi}$ takes a mix of positive and negative region proposals as input and outputs final classification probabilities $\boldsymbol{c}$, where $\boldsymbol{c}_i \in \mathbb{R}^{|C|+1}$ (with $|C|$

foreground classes plus one background class), as well as refined bounding box coordinates $\boldsymbol{l}$. Here, $|C|$ denotes the number of classes. The loss function for optimizing Faster R-CNN is defined as follows:

$$\mathcal{L}_{RPN}(p, p^*, \boldsymbol{l}, \boldsymbol{l}^*]) = \frac{1}{N_{cls}} \sum_i \mathcal{L}_{cls}(p_i, p_i^*) + \frac{1}{N_{reg}} \sum_i p_i^* \mathcal{L}_{reg}(\boldsymbol{l}_i, \boldsymbol{l}_i^*), \tag{1}$$

$$\mathcal{L}_{ROI}(\boldsymbol{c}, \boldsymbol{c}^*, \boldsymbol{l}, \boldsymbol{l}^*) = \frac{1}{N_{cls}} \sum_i \mathcal{L}_{cls}(\boldsymbol{c}_i, \boldsymbol{c}_i^*) + \frac{1}{N_{reg}} \sum_i p_i^* \mathcal{L}_{reg}(\boldsymbol{l}_i, \boldsymbol{l}_i^*), \tag{2}$$

$$\mathcal{L} = \mathcal{L}_{RPN} + \mathcal{L}_{ROI}, \tag{3}$$

where $\mathcal{L}_{cls}$ denotes the cross-entropy classification loss, $\mathcal{L}_{reg}$ denotes the smooth L_1 regression loss, $p_i^* \in \{0, 1\}$ for negative and positive anchors respectively, $\boldsymbol{c}_i^*$ and $\boldsymbol{l}_i^*$ denotes the classification labels and bounding box regression target, respectively.

3.2. Overall Framework

The overall framework of our proposed method, named GDA-IOD, is depicted in Fig. 2. This method is built upon a two-stage Faster-RCNN detector, incorporating a dynamic pseudo-label estimation strategy and an adaptive gradient alignment mechanism to mitigate catastrophic forgetting.

To mitigate forgetting caused by missing annotation while maximizing the utility of unlabeled old class objects. We take $\mathcal{M}_{t-1}$ to detect unannotated objects for each image in $\mathcal{D}_t$ and model the bimodal distribution often observed in inference results by Gaussian Mixture Model, named as GM-Pseudo. GM-Pseudo dynamically refines the predictions into pseudo labels and latent objects. We use pseudo labels for supervision training and exclude latent objects being misclassified as background in RPN. Based on estimated pseudo-labels, we further propose Gradient Decomposition and Alignment to dynamically balance model stability and plasticity. We propose a method to decompose the gradient $\boldsymbol{g}_{pseudo}$ associated with pseudo-labeling as a regularization constraint. When the angle between the overall gradient $\boldsymbol{g}$ and $\boldsymbol{g}_{pseudo}$ exceeds 90 degrees, we adjust $\boldsymbol{g}$ by projecting it orthogonally onto $\boldsymbol{g}_{pseudo}$ ensuring the two gradients align in the most effective direction. This gradient alignment strategy allows beneficial knowledge transfer between old and new classes.

3.3. Gaussian Mixture Model for Effective Pseudo Labeling

In this section, we propose using a Gaussian Mixture Model for refining the utilization of pseudo-labels. Missing annotations of previously learned class objects in current training images is a crucial problem in IOD, we first detect the old class objects in $\mathcal{D}_t$ by $\mathcal{M}_{t-1}$. Let $\mathcal{R}_t = \{(\boldsymbol{l}_i, \boldsymbol{c}_i, s_i)\}$ denote the inference result and $\mathcal{S}_t = \{s_i\}$, where $s_i \in [0, 1]$ represents the confidence score. As shown in Fig. 1, we leverage the fact that the confidence score set $\mathcal{S}_t$ tends to exhibit a double-peaked (bimodal) distribution, which can be categorized into three parts: **(1) low-confidence regions**, where predictions are mostly incorrect and should be discarded; **(2) medium-confidence regions**, where surpassing half objects are detected precisely but misclassified (false positive); and **(3) high-confidence regions**, where predictions are generally correct (true positive).

To reinforce the retention of prior knowledge by taking full use of unlabeled old class objects as supervision, we propose modeling the confidence scores as a bimodal distribution using a Gaussian Mixture Model (GMM) consisting of two Gaussian components with a **shared covariance matrix**. The GMM is defined as follows:

$$\mathrm{GMM}(s_i) = \pi_l \mathcal{N}(s_i | \mu_l, \sigma^2) + \pi_h \mathcal{N}(s_i | \mu_h, \sigma^2), \tag{4}$$

where $\mathcal{N}(s_i | \mu, \sigma^2)$ is a Gaussian distribution with mean μ and variance σ^2, μ_l and μ_h correspond to the means of the low-confidence and high-confidence regions, respectively, σ^2 is the shared variance.

Then, the Expectation Maximization (EM) algorithm is adopted to estimate the parameters of the $\mathrm{GMM}(\mu_l, \mu_h, \sigma^2, \pi_l, \pi_h)$ in an unsupervised learning manner. Upon convergence of the EM algorithm, we classify the inference results by confidence scores into three distinct regions based on the learned GMM parameters: (1) $s_i \leq \frac{\mu_l + \mu_h}{2}$, we discard these inferences, as they do not provide useful information. (2) $\frac{\mu_l + \mu_h}{2} < s_i \leq \mu_h - \alpha \cdot \sigma$, these are treated as potential targets, and the corresponding anchors are prevented from being selected as negative anchors in the Region Proposal Network. (3) $s_i > \mu_h - \alpha \cdot \sigma_h$, we take these subset of high-confidence inferences as $\mathcal{Y}_{old}$. Following [29], we further narrow $\mathcal{Y}_{old}$ by filtering out the one overlap with ground truth labels $\mathcal{Y}_t$ in $\mathcal{D}_t$ above an IoU threshold η_{iou} and take the rest as pseudo labels $\mathcal{Y}_{pseudo}$ for supervised training of current model $\mathcal{M}_t$.

With the addition of this threshold self-adjusting strategy, GM-Pseudo ensures adaptive stability in pseudo-label selection throughout the incremental training process.

3.4. Gradient Decomposition and Alignment

Although using pseudo-labels to optimize the model during incremental training helps mitigate catastrophic forgetting to some extent, the imbalance between old and new task samples in the dataset can still skew the model's predictions, reducing accuracy on previous tasks. To address this, we propose Gradient Decomposition and Alignment (GDA), a method that combines pseudo-labels with ground truth labels. GDA dynamically balances the model's stability and

plasticity by adaptively aligning gradients, improving performance across both new and old tasks.

Since adopting pseudo labels for joint optimization serves as a knowledge-replay mechanism in preserving old task knowledge, we propose the following hypothesis: if the loss related to old categories remains stable during incremental learning, the model effectively retains knowledge of previously learned tasks. Based on this hypothesis, we decompose a part of $\mathcal{L}_{ROI}$ in eq.(2) as $\mathcal{L}_{ROI}^{pseudo}$, which can be formulated as follows:

$$\begin{aligned}\mathcal{L}_{ROI}^{pseudo}(\boldsymbol{c}, \boldsymbol{c}^*, \boldsymbol{l}, \boldsymbol{l}^*) = \frac{1}{N_{cls}} \sum_i p_i^{'} \mathcal{L}_{cls}(\boldsymbol{c}_i, \boldsymbol{c}_i^*) \\ + \frac{1}{N_{reg}} \sum_i p_i^{'} \mathcal{L}_{reg}(\boldsymbol{l}_i, \boldsymbol{l}_i^*),\end{aligned} \tag{5}$$

where $p_i^{'} = 0$ when corresponding $\boldsymbol{c}_i^*$ is in $\mathcal{C}_i$ and $p_i^{'} = 1$ when corresponding $\boldsymbol{c}_i^*$ belongs to $\cup_{i=1}^{t-1}\mathcal{C}_i$. This means that $\mathcal{L}_{ROI}^{pseudo}$ only includes loss related to the categories from previous tasks. It's worth noting, we only decompose class-special $\mathcal{L}_{ROI}$, as $\mathcal{L}_{RPN}$ is class-agnostic in Faster RCNN. Accordingly, the optimization objectives based on the above hypothesis can be mathematically formulated as follows:

$$\text{Minimize}_\theta \ \mathcal{L} \qquad s.t. \ (\mathcal{L}_{ROI}^{pseudo})^{t-1} \geq (\mathcal{L}_{ROI}^{pseudo})^{t}, \tag{6}$$

where $(\mathcal{L}_{ROI}^{pseudo})^{t-1}$ and $(\mathcal{L}_{ROI}^{pseudo})^{t}$ represent $\mathcal{L}_{ROI}^{pseudo}$ before and after the parameter update. Since the model's parameters are updated through backpropagation based on the loss function, we can rephrase Eq. (6) as follows:

$$\text{Minimize}_\theta \ \mathcal{L} \qquad s.t. \left\langle \frac{\partial \mathcal{L}}{\partial \theta}, \frac{\partial \mathcal{L}_{ROI}^{pseudo}}{\partial \theta} \right\rangle \geq 0, \tag{7}$$

where $\langle , \rangle$ represents cosine similarity. Let $\boldsymbol{g}_{pseudo} = \frac{\partial \mathcal{L}_{ROI}^{pseudo}}{\partial \theta}$ and $\boldsymbol{g} = \frac{\partial \mathcal{L}}{\partial \theta}$, respectively. The essence of the above hypothesis is that if keep $\langle \boldsymbol{g}_{pseudo}, \boldsymbol{g} \rangle \geq 0$ in the incremental stage, parameter updating is unlikely to compromise knowledge of previous tasks and transfer beneficial knowledge to the previous task.

To meet derived bound in Eq. (7), we project $\boldsymbol{g}$ orthogonally onto $\boldsymbol{g}_{pseudo}$ when $\langle \boldsymbol{g}_{pseudo}, \boldsymbol{g} \rangle < 0$, which can be formulated as follows:

$$\boldsymbol{g}_{proj} = \begin{cases} \boldsymbol{g} & \text{if } \ \langle \boldsymbol{g}_{pseudo}, \boldsymbol{g} \rangle \geq 0 \\ \boldsymbol{g} - \lambda \cdot \frac{\boldsymbol{g}_{pseudo}^\top \cdot \boldsymbol{g}}{||\boldsymbol{g}_{pseudo}||^2} \cdot \boldsymbol{g}_{pseudo} & \text{otherwise,} \end{cases} \tag{8}$$

where λ is a hyperparameter that controls the strength of regularization. In this manner, we update the model parameters in the direction of the projected gradient vector $\boldsymbol{g}_{proj}$.

Notably, this orthogonal projection strategy does not prevent the model from learning new tasks. This can be theoretically shown, as $\langle \boldsymbol{g}_{proj}, \boldsymbol{g} \rangle \geq 0$ for any $\lambda \in [0, 1]$ (the detailed proof is available in Appendix), ensuring updates continue to drive learning forward. Our experiments further validate the effectiveness of GDA in reducing catastrophic forgetting and in mitigating overfitting due to imbalanced sample distributions between old and new tasks.

4. Experiments

4.1. Experiment Setup

Dataset and evaluation metrics. Our experiments are conducted on two widely used object detection benchmarks: Pascal VOC2007 [9] and MS COCO2017 [23]. Pascal VOC2007 includes 9,963 images with bounding box annotations for 20 distinct object categories. MS COCO2017 contains over 118,000 images with bounding box annotations for 80 object categories. In our experimental setup, "A-B" denotes that the initial task includes A categories, with each subsequent task introducing B additional categories. Following prior work [4, 28, 32, 34, 49], we organize the dataset by splitting categories in alphabetical order. For each image in dataset D_i, there is at least one object instance from category set C_i. Additionally, in line with previous works [28, 50, 56], we employ mAP across IoU thresholds ranging from 0.5 to 0.95, along with AP_{50} and AP_{75}, for evaluation in COCO-based settings, while the standard AP_{50} metric is used for evaluation in VOC-based settings. Our approach focuses on incremental object detection without relying on exemplar replay.

Implementation Details. We use Faster R-CNN [35] with ResNet-50 [15] backbone as the base object detector. The initial weight for backbone is pre-trained on ImageNet [7]. Stochastic gradient descent (SGD) with a momentum of 0.9 and weight decay of 1×10^{-4} is adopted as the optimizer. For incremental task training, the model is trained for 12 epochs with an initial learning rate of 2×10^{-2}, decaying by 0.1 and 0.01 at the 8th and 10th epoch in the first task, and for 6 epochs with an initial learning rate of 2×10^{-2}, decaying by 0.1 and 0.01 at the 3th and 5th epoch. The IoU threshold η_{iou} in Sec. 3.3 is set to 0.7. We set α in GM-Pseudo is set to 1.0 for VOC-based settings and 2.0 for COCO-based settings, respectively. λ in Eq. (8) to 0.6 to balance the strength of regularization.

4.2. Comparison with State-of-the-art Methods

To ensure fair comparison with previous studies, we evaluate our proposed method in both single-step and multi-step incremental learning scenarios. We establish two common baselines for reference: (1) **Joint Training**: where models are trained on all categories at once using a fully annotated dataset, and (2) **Fine-tuning**: where models are incrementally trained on each dataset $\mathcal{D}_i$ without any mechanisms to prevent catastrophic forgetting.

Single Step Incremental Settings. For Pascal

Table 1. Comparison with state-of-the-art methods in single-step incremental settings. We use **bold** to indicate the best results and underlines to highlight the second-best results. The methods that adopt example-replay (**rehearsal-based methods**) are marked by $*$ for fair comparison. Our method significantly outperforms state-of-the-art methods on all datasets.

Methods	Venues	10+10			15+5			19+1			5+15		
		Old	New	All	Old	New	All	Old	New	All	Old	New	All
Joint Training	-	77.2	76.4	76.8	78.3	72.5	76.8	76.8	77.4	76.8	74.7	77.5	76.8
Fine-tuning	-	9.6	67.4	38.5	11.2	63.4	24.3	9.4	72.5	12.6	0.3	75.4	56.6
ORE* [19]	CVPR21	60.4	68.8	64.6	71.8	58.7	68.5	69.4	60.1	68.9	-	-	-
ILOD-Meta* [20]	TPAMI21	68.4	64.3	66.3	71.7	55.9	67.8	70.9	57.6	70.2	-	-	-
OW-DETR* [13]	CVPR22	63.5	67.9	65.7	72.2	59.8	69.1	70.2	62.0	69.8	-	-	-
ABR* [28]	ICCV23	71.2	72.8	72.0	73.0	65.1	71.0	71.0	69.7	70.9	64.7	71.0	69.4
GMDP-ABR* [43]	ICLR25	72.1	73.2	72.7	75.8	**65.5**	73.2	74.8	**70.1**	74.6	67.1	71.9	70.7
Faster ILOD [34]	NeurIPS15	69.8	54.5	62.1	71.6	56.9	67.9	68.9	61.1	68.5	62.0	37.1	43.3
PPAS [55]	Arxiv20	63.5	60.0	61.8	-	-	-	70.5	53.0	69.2	-	-	-
MVC [49]	PR22	66.2	66.0	66.1	69.4	57.9	66.5	70.2	60.6	69.7	-	-	-
MMA [4]	CVPR22	69.3	63.9	66.6	73.0	60.5	69.9	71.1	63.4	70.7	66.8	57.2	59.6
ROSETTA [48]	CVPR22	67.6	66.0	66.8	71.7	61.6	69.2	69.7	68.3	69.6	-	-	-
PseudoRM [50]	MM23	69.1	68.6	68.9	73.4	60.9	70.3	72.9	67.3	72.6	-	-	-
PROB [56]	CVPR23	66.0	67.2	66.5	73.5	60.8	70.1	73.9	48.5	72.6	-	-	-
BPF [32]	ECCV24	71.7	74.0	72.9	75.9	63.0	72.7	74.5	65.3	74.1	66.4	75.3	73.0
GDA-IOD	-	**75.1**	**74.6**	**74.9**	**77.1**	63.2	**73.6**	**76.1**	58.4	**75.2**	**69.2**	**75.8**	**74.1**

Table 2. Comparison with state-of-the-art methods on MS COCO. The methods that adopt example-replay (**rehearsal-based methods**) are marked by $*$ for fair comparison.

Methods	40+40			70+10		
	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
Joint Training	40.2	61.5	44.2	40.2	61.5	44.2
Fine-tuning	21.6	33.4	23.1	5.6	8.6	6.2
ILOD-Meta* [20]	23.8	40.5	24.4	-	-	-
ABR* [28]	34.5	57.8	35.2	31.1	52.9	32.7
GMDP-ABR* [43]	**36.8**	**59.6**	36.7	32.5	53.8	33.9
Faster ILOD [34]	20.6	40.1	-	21.3	39.9	-
PseudoRM [50]	25.3	44.4	-	-	-	-
MMA [4]	33.0	56.6	34.6	30.2	52.1	31.5
BPF [32]	34.4	54.3	37.3	36.2	**56.8**	38.9
GDA-IOD	35.3	55.2	**38.3**	**36.5**	56.5	**39.8**

VOC2007, we sort categories alphabetically and add 1 (19-1), 5 (15-5), 10 (10-10), and 15 (5-15) classes in a single step. As shown in Tab. 1, **Fine-tuning** exhibits severe performance degradation on categories from previous tasks, with mAP on the old tasks dropping to just 0.3% in the 5-15 scenario. Our proposed GDA-IOD effectively mitigates catastrophic forgetting, showing consistent improvement across all settings. Notably, GDA-IOD surpasses BPF [32] by 3.4% and 0.6% mAP on old and new tasks under 10-10 respectively, indicating that GDA-IOD achieves better stability without sacrificing plasticity. For MS COCO2017, we report incremental results for the first 40 + last 40 and first 70 + last 10 incremental settings in Tab. 2. Enhanced by maximizing the utility of pseudo old-class objects as a knowledge-replay mechanism and preserving old task knowledge through gradient alignment, GDA-IOD consistently outperforms the rehearsal-free methods [4, 32]. Specifically, GDA-IOD achieves the highest AP, *e.g.* 35.3% and 36.5% under 40+40 and 70+10 scenarios, respectively.

Multiple Step Incremental Settings. We further report performance results under multi-step incremental settings. As shown in Tab. 3, ABR [28] set a strong baseline by adapting mixup and mosaic box augmentation strategies to replay previous objects. However, GDA-IOD demonstrates superior performance across most evaluation metrics, achieving an mAP of 63.5% on the challenging 5-5 setting and surpasses the rehearsal-based method ABR by over 7% mAP on the last 15 categories (6-20). We attribute this improvement to aligning the parameter update direction with the gradient preserving learned knowledge that does not hinder the model's plasticity, as mathematically proven in Sec. 3.4, while regularizing overfitting on new tasks.

4.3. Ablation Study

In the following, we conduct extensive ablation experiments to validate the effectiveness of the critical components of our proposed GDA-IOD.

Contributions from Different Components. As

Table 3. Comparison with state-of-the-art methods in multiple step incremental settings. The methods that adopt example-replay (rehearsal-based methods) are marked by $*$ for fair comparison.

Methods	Venues	Rehearsal	10+5+5(3 tasks)			5+5+5+5(4 tasks)			15+1+1+1+1+1(6 tasks)		
			1-10	11-20	1-20	1-5	6-20	1-20	1-15	16-20	1-20
Faster ILOD [34]	NeurIPS15	✗	68.3	57.9	63.1	55.7	16.0	25.9	66.9	44.5	61.3
MMA [4]	CVPR22	✗	66.7	61.8	64.2	62.3	31.2	38.9	68.3	54.3	64.1
ABR* [28]	ICCV23	✓	68.7	67.0	67.9	64.7	56.4	58.4	68.7	56.7	65.7
BPF [32]	ECCV24	✗	69.1	**68.2**	68.7	60.6	63.1	62.5	71.5	53.1	66.9
GMDP-ABR* [43]	ICLR25	✓	69.9	67.8	68.9	**66.3**	59.3	61.1	69.5	**58.9**	66.9
GDA-IOD	-	✗	**71.3**	67.2	**69.3**	62.2	**63.9**	**63.5**	**72.8**	50.4	**67.2**

Table 4. Performance comparison of different components in our method. Our model significantly improves over the baseline, especially in a relatively challenging setting.

Model	GM-Pseudo	GDA	VOC(5-5)					VOC(10-5)			
			1-5	6-10	11-15	16-20	1-20	1-10	11-15	16-20	1-20
(a)	✗	✗	57.0	53.7	68.1	55.2	58.5	69.4	73.6	60.1	68.1
(b)	✓	✗	61.0	58.4	68.9	58.6	61.7	70.8	73.0	59.8	68.6
(c)	✓	✓	**62.2**	**61.9**	**71.0**	**58.8**	**63.5**	**71.3**	**74.5**	**59.9**	**69.3**

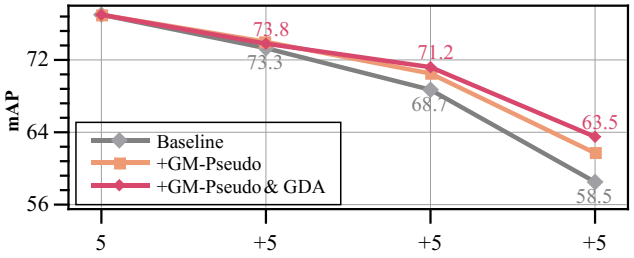


Figure 3. Task-wise average precision on test dataset under VOC 5-5 settings.

shown in Tab. 4, we conduct ablation studies by adding each component sequentially under VOC 10-5 and relatively challenging VOC 5-5 settings. First, we reproduce the model with the general pseudo-labeling method [29] as baseline model (a). The results Tab. 4 show that adaptively estimating pseudo-labels and employing a dynamic gradient alignment mechanism are effective in both incremental settings. By applying the Gaussian Mixture Model, GM-Pseudo not only takes the most confident object predictions as supervision but also maximizes the utility of pseudo-old class information by categorizing false positive inference results with medium confidence as latent objects. GM-Pseudo improves mAP by 3.2% under the 5-5 scenario. Moreover, compared to model (b), model (c) with GDA shows better performance on both old and new tasks. This indicates our proposed gradient alignment strategy effectively prevents the parameter updating comprising the learned knowledge from previous tasks, promoting beneficial knowledge transfer.

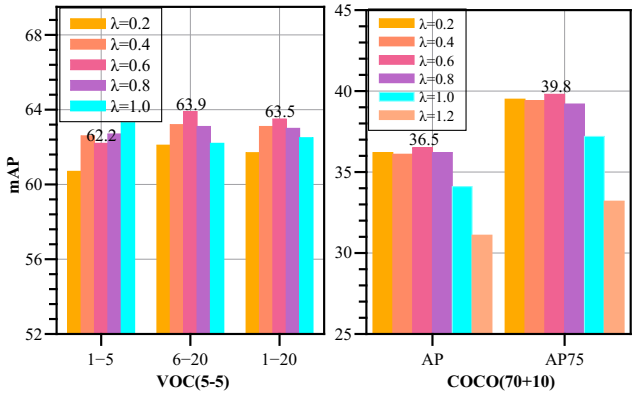


Figure 4. Ablation experiments with varying λ in Eq. (8) on VOC 5-5 and COCO 70-10 settings.

The issue of introducing pseudo-label noise is more crucial under longer incremental settings. Previous works [29, 32] set a fixed threshold to select pseudo labels, ignoring that the old model's performance is continually changing. As illustrated in Fig. 3, our proposed methods show accumulated performance improvement with the task increasing.

Hyper-parameter λ and α. The hyper-parameter λ in Eq. (8) is introduced to control the strength of regularization and we conduct ablation studies to analyze the impact of λ. As shown in Fig. 4, the performances achieve the best when $\lambda = 0.6$ on both VOC 5-5 and COCO 70-10. It's worth noting that performance drops dramatically when $\lambda > 1.0$ on COCO 70-10. We attribute this to $\lambda > 1.0$ preventing convergence (as analyzed in Sec. 3.4). Moreover, we also analyze hyper-parameter α in GM-Pseudo. As shown



Figure 5. Qualitative results for GM-Pseudo in the incremental stage under the 10-10 setting on Pascal VOC2007. The categories "chair," "car," and "cat" are old classes from the first stage without annotation. Compared with baseline, our proposed GM-Pseudo module adaptively takes predictions with medium confidence score as latent objects (**LO**), alleviating catastrophic forgetting caused by misclassifying this part of foreground targets as background.

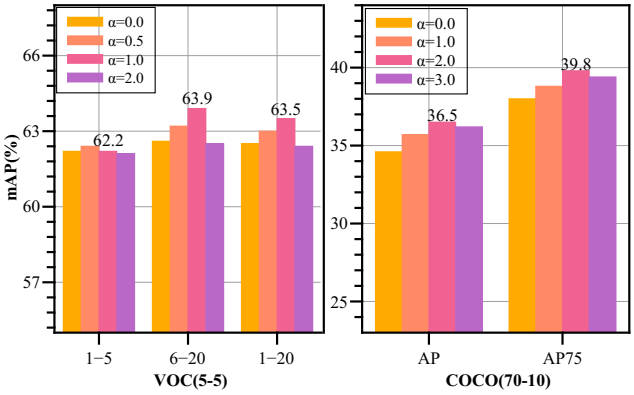


Figure 6. Ablation experiments with varying α in GM-Pseudo on VOC 5-5 and COCO 70-10 settings.

in Fig. 6, our model achieves the best performance when $\alpha = 1.0$ on VOC 5-5 and $\alpha = 2.0$ on COCO 70-10. We infer that COCO likely necessitates a larger α due to its more pronounced long-tail distribution, which requires maintaining relatively lower confidence pseudo-labels for difficult samples from previous classes.

4.4. Visulization

To better understand the effect of our proposed GM-Pseudo, Fig. 5 presents a comparison between GM-Pseudo and the baseline method. We empirically found that model $\mathcal{M}_{t-1}$ tends to produce lower confidence scores for some foreground instances, particularly in target-intensive scenarios. However, setting a relatively low confidence threshold for pseudo label selection is ineffective, which includes more false positives and false negatives predictions (as demonstrated in Tab. 1). In contrast, our proposed GM-Pseudo offers a more flexible and robust solution. It dynamically adjusts the thresholds based on the inference results and classifies medium-confidence predictions as latent objects. This approach ensures high-quality pseudo labels while preventing these medium-confidence regions from being misclassified as background in the Region Proposal Network.

5. Conclusion

This work addresses incremental object detection by tackling class overlap and enhancing knowledge retention. We propose an adaptive pseudo-labeling strategy using a Gaussian Mixture Model to identify the unlabeled objects. To further balance stability and adaptability, we introduce an adaptive gradient decomposition and alignment mechanism that facilitates beneficial knowledge transfer between old and new classes. Extensive experiments show that our approach effectively mitigates catastrophic forgetting while preserving high adaptability, making it a promising solution for dynamic, real-world object detection tasks.

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